

Rainfall Occurrence–Intensity Heterogeneity and Its Implications for Provincial Extreme Rainfall Design

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Abstract

Provincial drainage design often rests on one design rainfall, yet gauges within a province differ in how rain arrives as well as how much falls. Daily rainfall from 13 gauges in Uttaradit Province, 1981–2014, described each record by its annual total, wet-day count, and mean wet-day intensity, and gave design rainfall from annual maxima. Gumbel, generalized extreme value, and log-Pearson type III distributions were fitted at every gauge, the best by small-sample corrected Akaike information criterion carried forward, and percentile intervals obtained from 2000 bootstrap resamples with that distribution held fixed and with it reselected each time. Estimates differed between gauges by a factor of 3.83 at 2 years and 12.53 at 100 years, while the spread across the three candidates reached a median factor of 1.09 to 1.56. Mean wet-day intensity correlated with the 2-year estimate at 0.73 but 0.48 at 100 years. The median relative 95% bootstrap interval width at 100 years was 145.9% with the distribution fixed and 217.7% when reselected, an effect strongly station dependent, from 0.71 to 10.25 times the fixed width. Applying the network mean at 100 years left five gauges below their own estimate, the worst by 65%. Between-gauge differences therefore exceed the spread across candidates throughout, while long-return-period uncertainty depends on how firmly the distribution is identified at each gauge.

Keywords: Design rainfall, Frequency analysis, Rainfall intensity, Return period, Uttaradit

1 INTRODUCTION

Culverts, roadside drains, farm ponds, and small-scale irrigation structures are sized from a design rainfall: the rainfall depth with an annual exceedance probability of $1/T$, corresponding to a return period of T years. This quantile forms the cornerstone of flood mitigation and disaster risk management, directly governing the structural safety and failure risk of local water infrastructure. Where a project lies away from a gauge, or where a program covers many sites at once, a single value is often adopted for an administrative area and applied throughout it. Whether the resulting value represents any particular site is less often examined.

Two things can make a provincial value unrepresentative, and they are not the same. The first is that rainfall extremes genuinely differ from place to place, so no single number fits everywhere. The second is that the estimate at any one gauge is itself uncertain, because a return level far beyond the length of the record is an extrapolation from a small sample. For a design rainfall at a long return period, two distinct sources of variation are therefore relevant: spatial variation among gauges, and the estimation uncertainty that attends extrapolation beyond the observed record [1].

Rainfall records also differ in a way that is easy to overlook. Two gauges can receive nearly the same annual total while one delivers it in 80 wet days and the other in 220, so that the mean depth on a wet day differs by a factor of two; indices describing this structure are standard in the analysis of daily records [2], [3]. This occurrence–intensity structure is a property of the record before any distribution is fitted, and it is natural to ask whether it is associated with the design estimate. If a strong association exists, this structure could provide a simple network-level descriptor; if not, the extremes must be estimated on their own terms.

Precipitation extremes over Thailand have been examined in studies of long-term trend and variability [3], [4], and comparable analyses cover mainland Southeast Asia [5] and the global land surface [6]; the literature on fitting and comparing extreme-value distributions is extensive [7]–[9]. Recent regional frequency analyses in Thailand have demonstrated substantial spatial heterogeneity in extreme rainfall and have explored regionalization using L-moments and homogeneous-region approaches [10], and Thailand-wide studies have developed regional intensity–duration–frequency relationships from rainfall records, providing a broader engineering context for site-level design estimation [11]. Which distribution best describes daily rainfall maxima has been surveyed globally without a single answer emerging [12], and the uncertainty attached to a long return level has been argued to dominate the choices made in fitting it [1]. This study compares, on a common relative scale, three quantities relevant to design practice within a

single administrative area: how far the estimate moves between gauges, how far it moves when a different distribution is fitted to the same gauge, and how far it moves under resampling of the same record.

This study examines three aspects of provincial extreme-rainfall design. First, it quantifies the differences in design rainfall among gauges within the province and evaluates how these differences vary with return period. Second, it assesses the association between the occurrence–intensity structure of rainfall records and design rainfall estimates across return periods. Third, it quantifies the representation error associated with applying a single provincial design value at individual gauges and compares this error with the uncertainty of gauge-specific estimates. All 13 gauges are considered in the primary analysis, with sensitivity analyses providing practical insight into the implications of spatial variability for provincial and site-specific rainfall design.

2 MATERIALS AND METHODS

2.1 Study area and data

Uttaradit Province lies in northern Thailand between about 17.2° and 18.1° N and 100.0° and 101.1° E. Daily rainfall for 1 January 1981 to 31 December 2014 was obtained from the Royal Irrigation Department and the Thai Meteorological Department for the 13 gauges operated within the province, at elevations from 55 to 428 m above mean sea level. Depths are recorded in millimeters to one decimal place as daily totals. Every gauge carries a numeric value on all 12,418 days of the period as supplied, with no missing-value codes and no negative entries, so record length and completeness are identical across the network and cannot contribute to any difference reported below. No additional quality control was applied here beyond that carried out by the supplying agencies, which is not documented in the delivered files; the three diagnostics of Section 2.11 examine the records for the specific artifacts they can detect. Table 1 gives the station characteristics.

Table 1: Characteristics of the 13 rain gauges in Uttaradit and the occurrence–intensity description of their records for 1981–2014. No. is the analysis number used throughout and Station is the operating agency's code. P , W , and S are defined by Equations (1) to (3), and the last column is the largest daily depth in the record.

No. Station	Lat. (°N)	Long. (°E)	Elev. (m)	P (mm)	W (d)	S (mm)	Max. daily (mm)
1 351001	17.23	100.1	55	997	80	12.37	259.6
2 351002	17.42	100.13	62	984	81	12.03	170
3 351003	17.67	100.68	372	1376	123	11.09	155
4 351004	17.65	100.38	116	1100	84	13.06	181.3
5 351005	17.6	100.05	81	1106	96	11.42	152
6 351006	17.92	100.88	349	1147	175	6.36	143.5
7 351007	18.02	101.07	428	1203	182	6.45	201.1
8 351008	17.47	100.37	82	1134	178	6.17	176.6
9 351009	17.75	100.28	95	1206	229	5.11	68
10 351010	17.9	100.82	215	1177	220	5.18	54.2
11 351011	17.56	100.11	65	1108	90	12.13	296.9
12 351012	17.75	100.92	415	944	79	11.83	298.5
13 351201	17.62	100.1	67	1065	93	11.35	226

2.2 Description of each record

A wet day is a day with at least 1.0 mm, applied identically at every gauge. Equations (1) to (3) describe the record before any distribution is fitted. The mean annual total is

$$P = \frac{1}{N_y} \sum_{t=1}^n x_t \quad (1)$$

where x_t is the depth on day t , n the number of days in the record, and N_y the number of years. The mean number of wet days per year is

$$W = \frac{1}{N_y} \sum_{t=1}^n I(x_t \geq 1.0) \quad (2)$$

with $I(\cdot)$ the indicator function. The simple daily intensity index is the mean depth on those days,

$$S = \frac{\sum_{t=1}^n x_t I(x_t \geq 1.0)}{\sum_{t=1}^n I(x_t \geq 1.0)} \quad (3)$$

Because Equation (1) counts every day while Equations (2) and (3) count only days at or above the 1.0 mm threshold, P is close to but not exactly equal to $W \times S$: across these 13 records the product falls short of the annual total by 0.58% to 3.31%, the difference being the rain that falls on days below the threshold. These three form the occurrence–intensity description referred to throughout, and Equation (3) follows the definition of the Expert Team on Climate Change Detection and Indices [2].

2.3 Annual maximum series

For each gauge and calendar year the annual maximum daily depth is

$$M_y = \max\{x_t : t \in \text{year } y\} \quad (4)$$

A year was retained only if it contained at least 300 days with a value, which every year at every gauge satisfies, giving 442 station-years from 13 gauges over 34 years. Equation (4) is the block maximum on which Equations (5) to (7) are fitted.

2.4 Distributions fitted

Three distributions in common engineering use [13] were fitted to each annual maximum series by maximum likelihood. The Gumbel distribution [7] has cumulative function

$$F(x) = \exp\left\{-\exp\left[-\frac{x-\mu}{\sigma}\right]\right\} \quad (5)$$

with location μ and scale σ . The generalized extreme value (GEV) distribution [8] adds a shape parameter ξ ,

$$F(x) = \exp\left\{-\left[1 + \xi \frac{x-\mu}{\sigma}\right]^{-1/\xi}\right\}, \quad 1 + \xi \frac{x-\mu}{\sigma} > 0 \quad (6)$$

and reduces to Equation (5) as $\xi \rightarrow 0$. The log-Pearson type III (LP3) distribution is a Pearson type III distribution fitted by maximum likelihood to the logarithms $z = \ln x$, in the skewness, location, and scale parameterization of the SciPy implementation used here. Because the fit is on the logarithmic scale, the log-likelihood on the scale of x carries the Jacobian of the transformation,

$$\ln f_X(x) = \ln f_Z(\ln x) - \ln x \quad (7)$$

which is the form entered into Equation (8). Return levels are obtained on the logarithmic scale and exponentiated back.

The three were compared at each gauge using the Akaike information criterion corrected for small sample size (AICc) [14], [15],

$$AICc = 2k - 2 \ln L + \frac{2k(k+1)}{n-k-1} \quad (8)$$

where k is the number of fitted parameters, two for Equation (5) and three for the other two, $n = 34$ the number of annual maxima, and L the maximized likelihood, summed over the sample using Equation (7) for the log-Pearson type III so that all three are compared on the scale of x . The correction was retained because the sample contains only 34 annual maxima, a size at which the uncorrected criterion tends to favor the more complex model; in this dataset, however, it did not change the selected distribution at any gauge. The uncorrected criterion was computed as an internal check and is not tabulated. Two further quantities describe how firmly the selection is made: $\Delta AICc$, the gap between the selected model and the next best, and the Akaike weight, $\exp(-\Delta AICc/2)$ normalized across the three candidates, which is the relative support each carries. The Kolmogorov–Smirnov statistic was also computed as a descriptive goodness-of-fit diagnostic,

$$D = \max_x |F_n(x) - F(x)| \quad (9)$$

with F_n the empirical distribution function; because F_n is a step function on a finite sample the supremum is attained, so the maximum in Equation (9) is exact. Because the parameters are estimated from the same sample,

Equation (9) is reported as a statistic only and no p-value is attached to it. The distribution with the lowest value of Equation (8) was carried forward at each gauge; the other two are retained to quantify how much the choice matters.

2.5 Return levels and their uncertainty

The design rainfall for return period T is the quantile with an annual exceedance probability of $1/T$,

$$R(T) = F^{-1}\left(1 - \frac{1}{T}\right) \quad (10)$$

evaluated for $T = 2, 5, 10, 25, 50$, and 100 . Two bootstraps were run, each with $B = 2000$ replicates in which the annual maxima are resampled with replacement to the original length n and Equation (10) re-evaluated [16]. They differ in what is held fixed. Both resample the annual maxima as if they were independent, so the intervals are conditional on that assumption; Section 3.8 reports where it is least comfortable. The *fixed-selection* bootstrap refits the distribution chosen from the original record, so its interval measures resampling variability conditional on that model choice. The *selection-inclusive* bootstrap refits all three candidates within every replicate and reselects by Equation (8), so its interval also incorporates variability arising from model reselection within the candidate set. Reporting both separates sampling variability from the additional contribution of choosing a model, which the first bootstrap cannot show. The interval is the 2.5th and 97.5th percentiles of the resulting distribution, and its width is reported relative to the point estimate,

$$w(T) = \frac{R_{97.5}(T) - R_{2.5}(T)}{R(T)} \times 100 \quad (11)$$

so that a value of 100% means the interval spans a range equal to the estimate itself. Equation (11) defines the quantity referred to as interval width throughout; it is a full width and not a half-width, and it measures the spread of the estimate under resampling, not the error of a prediction.

2.6 Comparison of three sources of spread on a common scale

Three ratios are formed from the same fitted estimates so that they can be compared directly. Between gauges,

$$\rho_g(T) = \frac{\max_i R_i(T)}{\min_i R_i(T)} \quad (12)$$

over the 13 gauges i . Within a gauge, across the three distributions d ,

$$\rho_d(T) = \text{median}_i \left[\frac{\max_d R_{i,d}(T)}{\min_d R_{i,d}(T)} \right] \quad (13)$$

and for the resampling spread the median of Equation (11) across gauges. The three are not components of a variance decomposition and are not additive; they are placed on a common relative scale so that their magnitudes can be compared.

2.7 Association between record structure and design estimate

Whether the occurrence–intensity descriptors were associated with the design estimate was assessed using Spearman rank correlations between each of P , W , and S and each of $R(T)$, $w(T)$, and the shape parameter ξ of Equation (6). Because the selected fits mix three parameters that are not comparable, the shape of the generalized extreme value distribution fitted at every gauge is used, whether or not that distribution was the one selected there. With ranks a_i and b_i of the two variables at gauge i ,

$$r_s = 1 - \frac{6 \sum_{i=1}^n (a_i - b_i)^2}{n(n^2 - 1)} \quad (14)$$

for $n = 13$. A rank correlation was used because no linear form is assumed. With 13 gauges these correlations are descriptive, and they are reported as associations rather than as evidence of a mechanism.

2.8 Provincial representation error

A single provincial value $R_p(T)$ was formed in four ways from the 13 gauge estimates: their mean, median, 75th percentile, and 90th percentile. The error it introduces at gauge i is expressed against that gauge's own estimate, which is the value a designer at that site would otherwise use,

$$E_i(T) = \frac{R_p(T) - R_i(T)}{R_i(T)} \times 100 \quad (15)$$

A negative value means the provincial figure falls below the local estimate, so a structure sized on it would be underdesigned relative to the local record; a positive value means the opposite. The two are counted separately because they carry different consequences.

2.9 Diagnostics of the annual maximum series

Fitting a stationary distribution to annual maxima assumes the values are drawn independently from one distribution throughout the record. Two diagnostics test that assumption directly. Serial dependence was examined by the lag-one autocorrelation of each series and by the Ljung–Box statistic over five lags. Temporal stability was examined by the Mann–Kendall test with the Theil–Sen slope, applied to the annual maxima rather than to the daily record. Both are diagnostics on the assumption, not a change of model: no nonstationary or dependent frequency model is fitted here, and the results are used to qualify how the return levels should be read. Statistical significance is assessed throughout at the two-sided 5% level.

2.10 Software and reproducibility

All computation used Python 3.12.3 with NumPy 2.4.4, SciPy 1.17.1, pandas 3.0.2, Matplotlib 3.10.8, statsmodels 0.15.0, and pymannkendall 1.4.3. The three distributions were fitted with `'scipy.stats.gumbel_r'`, `'scipy.stats.genextreme'`, and `'scipy.stats.pearson3'`. SciPy parameterizes the generalized extreme value distribution with a shape c related to the ξ of Equation (6) by $\xi = -c$, verified by simulation, and the sign convention of Equation (6) is used wherever a shape parameter is reported. Both bootstraps drew from `'numpy.random.default_rng'` with a fixed seed, 0 for the fixed-selection bootstrap and 2024 for the selection-inclusive one, so the intervals reproduce exactly. The analysis scripts, the configuration file, and the result tables underlying every number in this paper are available from the corresponding author on request.

2.11 Scenarios and diagnostic checks

Because two gauges have a markedly lower S than the rest, three scenarios are reported: all 13 gauges, the two lowest- S gauges set aside, and the five lowest set aside. These are scenarios and not corrections. No threshold on S is proposed, and no claim is made that any subset gives a better estimate.

Three diagnostic checks were applied to the daily records to test whether an unusual occurrence–intensity structure is accompanied by signs of a recording artifact. The first is the frequency of terminal digits among wet-day values, where rounding to coarser increments can leave an excess of values ending in zero or five. The second is a chi-square test of wet-day counts by day of the week, where an observation routine tied to the calendar would leave a pattern. The third takes the 50 days on which the other 12 gauges were wettest on average and asks what the gauge in question recorded, which distinguishes a record that misses the network's rain events from one that registers them at lower depths.

3 RESULTS AND DISCUSSION

3.1 Occurrence–intensity structure of the network

The 13 records differ far more in how rainfall arrives than in how much of it falls (Table 1, Figure 1). Mean annual totals span 944 to 1,376 mm, a factor of 1.46. Over the same network the mean number of wet days spans 79 to 229 per year, a factor of 2.91, and the mean wet-day intensity spans 5.11 to 13.06 mm, a factor of 2.56. Within this network the two vary inversely to a degree that largely offsets them: the rank correlation between wet-day count and S is -0.978 , so a record with many wet days tends to have correspondingly light ones and reaches a similar annual total. This is a property of these 13 records and is not asserted as a general one.

The largest daily depth in each record spans 54.2 to 298.5 mm, a factor of 5.51, varying far more widely across the network than the annual total; it is shown against mean wet-day intensity in Figure 1c. Two gauges, 351009 and 351010, recorded no day above 100 mm in 34 years while receiving annual totals of 1,206 and 1,177 mm; a third of the network recorded such a day more than twenty times.

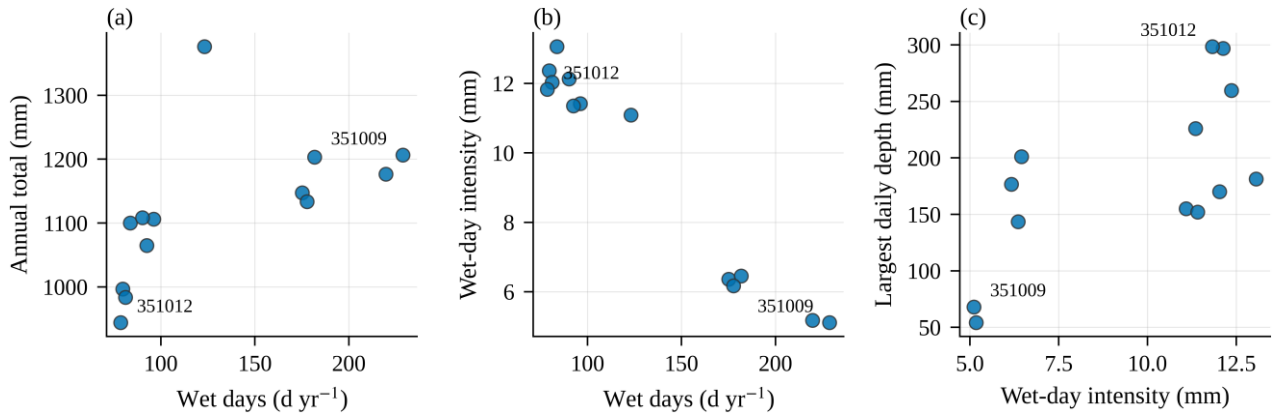


Figure 1: Occurrence–intensity structure of the 13 records: (a) mean annual total against mean number of wet days, (b) mean wet-day intensity against mean number of wet days, and (c) the largest daily depth in the record against mean wet-day intensity. The two gauges named in Section 3.1 are labeled; the remaining codes are in Table 1. The near-perfect compensation between wet-day count and intensity in (b) helps explain why the annual totals in (a) span a much narrower range than either component.

3.2 Distributions selected

Equation (8) selected the log-Pearson type III at six gauges, the Gumbel at five, and the generalized extreme value at two (Table 2). The selections were identical to those obtained with the uncorrected criterion at all 13 gauges, so the small-sample correction did not change the distribution chosen anywhere in the network.

The selections were nevertheless weakly separated. The Akaike weight of the selected model ranged from 0.43 to 0.99 with a median of 0.54, and at 11 of the 13 gauges the second-ranked model lay within 2 AICc units of the selected one. Gauge 351009 is representative: log-Pearson type III carried a weight of 0.456 and the generalized extreme value 0.397, separated by 0.28 AICc units. Gauge 351003 was the only one with a clearly dominant selected model, carrying a weight of 0.990 and a gap of 9.65 units. The modal distribution at a gauge should therefore not be read as an unambiguous model choice, and the counts of six, five, and two describe which model ranked first rather than which one the record identifies. Kolmogorov–Smirnov statistics were also computed as descriptive diagnostics and are not used to select or corroborate the selection.

Table 2: AICc values for the three candidate distributions fitted to the annual maximum daily rainfall at each gauge, together with the Δ AICc to the second-ranked model, the Akaike weight of the selected model, and the selected distribution.

Station	AICc Gumbel	AICc GEV	AICc LP3	Δ AICc	Akaike weight	Selected
351001	354.4	351.4	351.1	0.3	0.488	LP3
351002	320.4	322.8	322.8	2.39	0.624	Gumbel
351003	355.4	353.1	343.4	9.65	0.99	LP3
351004	340.1	341.7	340.8	0.62	0.454	Gumbel
351005	320.4	322.3	322.1	1.64	0.543	Gumbel
351006	309.9	291.3	291.6	0.31	0.539	GEV
351007	327.7	303.3	304.1	0.76	0.594	GEV
351008	333.9	333.2	332.4	0.82	0.473	LP3
351009	251.8	249.8	249.5	0.28	0.456	LP3
351010	249.3	250.3	249.9	0.59	0.426	Gumbel
351011	376.5	375.1	373.4	1.69	0.609	LP3
351012	365.4	436.3	366	0.57	0.57	Gumbel
351201	342.2	339.1	338.6	0.52	0.518	LP3

3.3 Design rainfall differs more between gauges than between distributions

Between-gauge differences are large and grow with return period (Table 3, Figure 2). The ratio of Equation (12) is 3.83 at a 2-year return period, 5.56 at 10 years, 7.67 at 25 years, and 12.53 at 100 years, the estimates spanning 57.8 to 724.7 mm at the longest period.

Against this, the spread across the three candidate distributions moves a gauge's own estimate by a median factor of 1.09 at 2 years and 1.56 at 100 years by Equation (13). This is the full range among the candidates, not the uncertainty of choosing among them, which Section 3.5 addresses separately. The spread between gauges therefore exceeds the spread across candidate distributions by a factor of about eight at 100 years and about three and a half at 2 years. Within this network and under these three distributions, where the gauge is located matters considerably more for the design estimate than which of the three distributions is fitted to it. The finding is specific to the three distributions compared here and to this network; a wider set of candidate distributions, or a network with more uniform records, need not behave the same way.

Table 3: Design rainfall and the sources of spread of Section 2.6. Upper block, by return period: smallest and largest gauge estimate, their ratio by Equation (12), the median within-gauge ratio across the three candidates by Equation (13), and the median relative interval width of Equation (11) as a percentage of the estimate under the fixed-selection and selection-inclusive bootstraps. Ratio range is the selection-inclusive width divided by the fixed-selection width, given as the range across the 13 gauges rather than its median, because the effect of reselection differs greatly between them. Lower block: the range and network mean at 25 and 100 years under the three scenarios of Section 2.11, with n the number of gauges retained; the scenarios change only which gauges enter the range and are a sensitivity, not a correction.

T (yr)	Min. (mm)	Max. (mm)	Gauge ratio	Dist. ratio	Fixed (%)	Selection (%)	Ratio range	Scenario	n
2	26.1	99.9	3.83	1.09	30.8	39.6	1.00–2.09		
5	35.3	153.8	4.35	1.01	38.7	40.4	0.98–1.24		
10	40.8	226.6	5.56	1.08	44.5	57.6	0.93–1.54		
25	47.7	365.8	7.67	1.25	75	113	0.85–3.65		
50	52.8	517.2	9.8	1.39	108.4	160.8	0.78–6.17		
100	57.8	724.7	12.53	1.56	145.9	217.7	0.71–10.25		
25	47.7	365.8	7.67					A	13
100	57.8	724.7	12.53					A	13
25	118.2	365.8	3.1					B	11
100	154.1	724.7	4.7					B	11
25	132.2	365.8	2.77					C	8
100	154.1	724.7	4.7					C	8

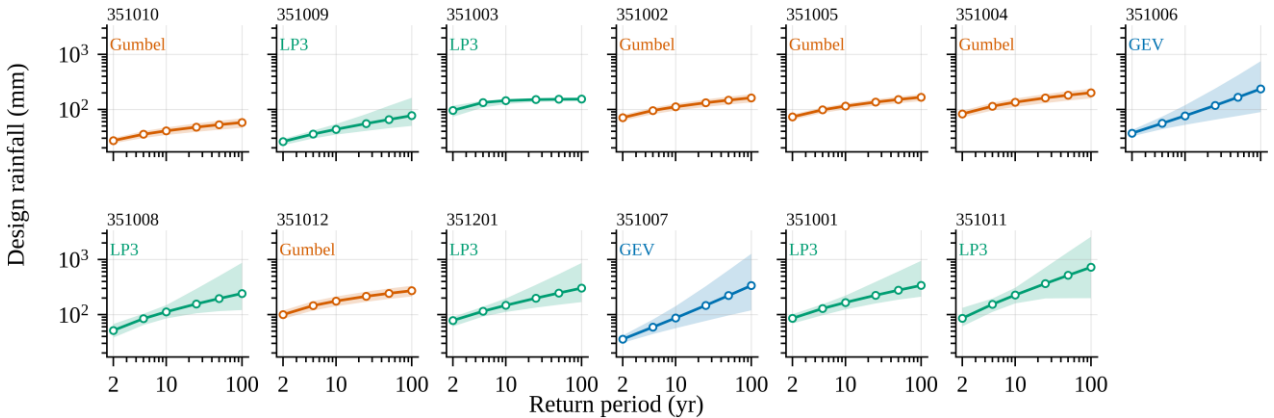


Figure 2: Design rainfall at each gauge for return periods from 2 to 100 years, with the 95% fixed-selection bootstrap interval of Section 2.5 shown as a shaded band; the selection-inclusive interval is not drawn here and is summarized in Table 3 and Figure 3. Color denotes the distribution selected at that gauge, vermilion for the Gumbel, blue for the generalized extreme value, and green for the log-Pearson type III, and the same colors are used for these distributions throughout. Gauges are ordered by their 100-year estimate. Panels share the same axis limits, and both axes are logarithmic because the estimates span more than one order of magnitude at the longest return period.

3.4 The record structure is less strongly associated with long-period estimates

Mean wet-day intensity is positively associated with the design estimate and wet-day count negatively so, as the compensating relation of Section 3.1 would suggest, but the associations generally weaken as the return period increases (Table 4). The Spearman correlation of Equation (14) between S and the return level is 0.73 at 2 years, 0.70 at 25 years, and 0.48 at 100 years; for wet-day count it is -0.80 , -0.70 , and -0.45 over the same range. Mean annual total shows the weakest of the three associations throughout, between -0.40 and -0.54 .

Two further associations are absent altogether. None of the three is associated with the width of the bootstrap interval, the correlations lying between -0.19 and $+0.19$ at every return period, nor with the shape parameter of the generalized extreme value distribution fitted at every gauge, where the correlations run from -0.22 to 0.06 and none reaches significance at the 5% level ($p = 0.47$ to 0.86).

The gauge-level values are consistent with this. Gauge 351009, which has the lowest S in the network at 5.11 mm, does not give the lowest 100-year estimate: at 77.2 mm it sits above 351010 at 57.8 mm, whose S is marginally higher. Gauge 351003, with an S of 11.09 mm in the middle of the upper group, gives a 100-year estimate of 154.1 mm, lower than gauge 351007 at 337.4 mm whose S is 6.45 mm. A low mean wet-day intensity therefore does not imply a low design rainfall. The occurrence–intensity descriptors were more strongly associated with shorter return-period estimates than with the upper tail, which is the part design depends on. These are rank associations over 13 gauges, computed without adjustment for the number of correlations examined, and they are not evidence that one quantity determines the other.

Table 4: Spearman rank correlations of Equation (14) between the occurrence–intensity description of a record and three properties of its fitted distribution, over the 13 gauges. Correlations with the shape parameter refer to the generalized extreme value shape ξ of Equation (6) fitted independently at all 13 gauges, irrespective of whether that distribution was the one selected there, because the shape parameters of the three candidates are not comparable with one another.

Property	T (yr)	P	W	S
Return level R(T)	2	-0.489	-0.802	0.731
	5	-0.473	-0.769	0.736
	10	-0.544	-0.775	0.736
	25	-0.516	-0.698	0.703
	50	-0.412	-0.527	0.533
	100	-0.396	-0.451	0.484
Interval width w(T)	2	-0.027	-0.269	0.264
	5	0.055	0.088	-0.055
	10	0.038	0.104	-0.055
	25	0.093	0.192	-0.154
	50	0.082	0.17	-0.115
	100	0.077	0.176	-0.121
GEV shape ξ	–	-0.22	-0.165	0.055

3.5 Resampling spread of the return-level estimates

Under the fixed-selection bootstrap the median interval of Equation (11) spans 30.8% of the estimate at a 2-year return period, 44.5% at 10 years, 75.0% at 25 years, and 145.9% at 100 years (Table 3, Figure 3). At 100 years the width exceeds 100% at seven of the 13 gauges and ranges from 12.5% to 343.6% across the network.

At the network median, allowing the distribution to be reselected within each replicate widens the interval further, but the network median masks substantial between-gauge heterogeneity. The median selection-inclusive width is 39.6% at 2 years, 57.6% at 10 years, 113.0% at 25 years, and 217.7% at 100 years, and it exceeds 100% at ten of the 13 gauges rather than seven. The ratio of the selection-inclusive width to the fixed-selection width at 100 years runs from 0.71 to 10.25 across the network while its median is only 1.11, demonstrating substantial between-gauge heterogeneity in the effect of model reselection on interval width.

The pattern follows the weight of evidence reported in Section 3.2. The widest increases occur where the selection is least firm. At gauge 351012 the Gumbel carries an Akaike weight of 0.57 against 0.43 for the log-Pearson type III, and across replicates the two were reselected in 51.1% and 48.4% of cases; its interval widens from 44.3% to 454.0%. Gauge 351004 behaves the same way and widens from 39.2% to 318.1%. At gauges 351006 and 351007 the selection-

inclusive width is smaller than the fixed-selection width, decreasing from 281.5% to 199.1% and from 343.6% to 243.0% respectively. The modal bootstrap choice matched the full-record choice at 12 of the 13 gauges, the exception being 351004, where the log-Pearson type III was reselected in 52.3% of replicates against 46.9% for the Gumbel chosen from the full record. The reselection shares and the Akaike weights quantify different aspects of model-selection stability: the former across resamples and the latter within the original record.

Both the between-gauge variation and the resampling spread are substantial at long return periods, and it is worth being precise about what the latter means. It is the spread of the estimate under resampling of the same 34 annual maxima, computed by Equation (11); it is a resampling spread, not a prediction error, not a bias, and not a total uncertainty: parameter uncertainty beyond resampling, distributions outside the three compared here, nonstationarity, and regionalization all lie outside what it measures. Within this study it says that a 100-year value obtained from 34 years at one gauge is poorly determined, and that reporting it without an interval conveys a precision the record does not support.

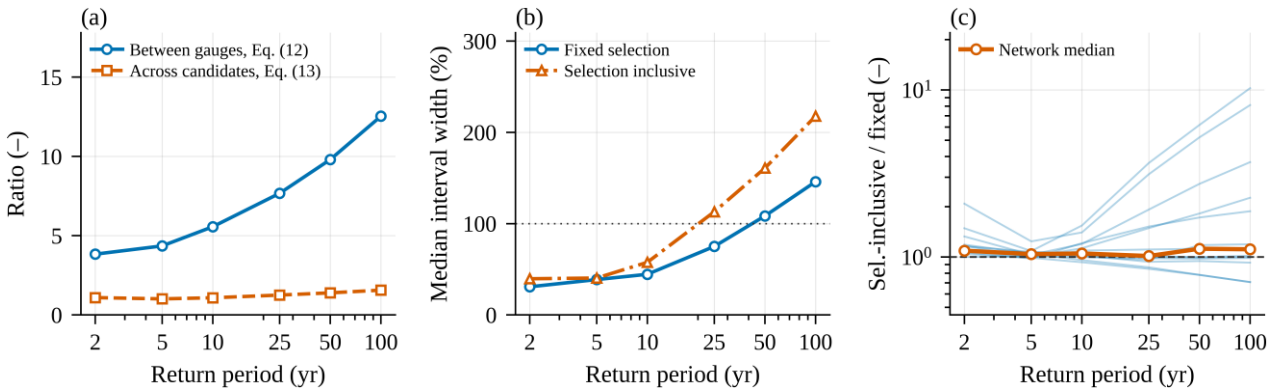


Figure 3: Comparison of spatial, distributional, and resampling spread across return periods. (a) The between-gauge ratio of Equation (12) and the median within-gauge ratio across the three candidate distributions of Equation (13). (b) The median relative interval width of Equation (11) under the fixed-selection and selection-inclusive bootstraps, with the dotted line marking a width equal to the estimate. (c) The ratio of the two widths at every gauge, in pale blue, with the network median in vermilion; the logarithmic scale is needed because the ratio spans more than an order of magnitude at 100 years. These quantities are not additive components of a variance and are placed on comparable scales only so that their magnitudes can be read against one another.

3.6 Representation error from a single provincial design value

Adopting the network mean as a provincial value at a 100-year return period gives 251.8 mm. By Equation (15) this falls below the local estimate at five gauges, by as much as 65.3%, and exceeds it at eight, by as much as 335.3%, with a median absolute error of 25.8% (Table 5). At a 25-year return period the corresponding figures are 162.2 mm, four gauges below by up to 55.7%, nine above by up to 240.2%, and a median absolute error of 22.7%.

The other three provincial statistics trade one error against the other rather than reducing both (Figure 4). The median reduces the largest overdesign to 308.6% but increases the number of gauges left below their local estimate from five to six. The 90th percentile reduces the count below to two and the worst shortfall to 53.2%, but raises the largest excess to 486.1% and the median absolute error to 53.2%. No choice of provincial statistic removes both kinds of error, because the underlying gauge estimates span more than an order of magnitude at this return period: any single value must fall below some of them and above the rest. Section 3.5 puts this in context: at 100 years the median representation error of 25.8% is substantially smaller than the median resampling spread of 145.9%, although the two are different quantities and are not components of a common uncertainty budget.

Table 5: Provincial representation error of Equation (15) at 25- and 100-year return periods, for four ways of forming a single provincial value. Gauges are counted separately according to whether the provincial value falls below or above their own estimate, and the worst case of each is reported.

T (yr)	Statistic	Value (mm)	Gauges below	Gauges above	Worst below (%)	Worst above (%)	Median error (%)
25	Mean	162.2	4	9	-55.7	240.2	22.7
25	Median	151.2	6	6	-58.7	217.1	23.9
25	75th percentile	198.7	3	9	-45.7	316.9	35.3

T (yr)	Statistic	Value (mm)	Gauges below	Gauges above	Worst below (%)	Worst above (%)	Median error (%)
25	90th percentile	222.4	2	11	-39.2	366.5	47.1
100	Mean	251.8	5	8	-65.3	335.3	25.8
100	Median	236.3	6	6	-67.4	308.6	30.4
100	75th percentile	302.8	3	9	-58.2	423.5	50.6
100	90th percentile	339	2	11	-53.2	486.1	53.2

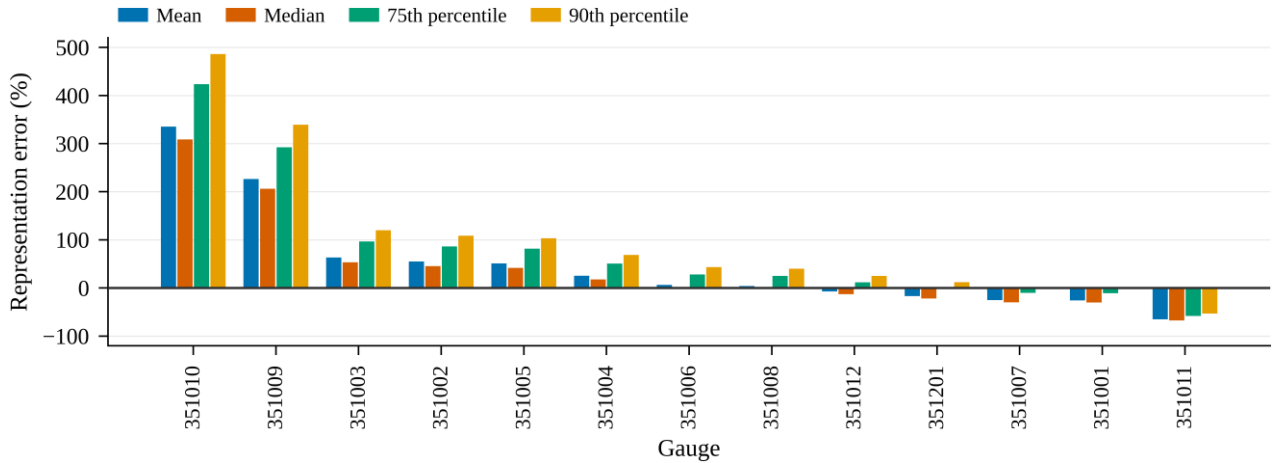


Figure 4: Provincial representation error of Equation (15) at each gauge for a 100-year return period, for the four provincial statistics of Section 2.8. Bars below zero mark gauges where the provincial value falls short of the local estimate and bars above it gauges where the provincial value exceeds it. The vertical scale is symmetric-logarithmic so that the smaller errors remain legible beside the two largest. No single provincial statistic removes both kinds of error.

3.7 Setting gauges aside changes the range but is not a correction

Setting aside the two lowest- S gauges reduces the between-gauge ratio at 100 years from 12.53 to 4.70 and raises the network mean from 251.8 to 285.3 mm. Setting aside five gauges gives exactly the same ratio, 4.70, and a mean of 290.3 mm, because the extreme estimates in the reduced network come from the same two gauges in both scenarios (Table 3, lower block).

That the second exclusion changes nothing is informative. It shows that a criterion based on S does not isolate the gauges responsible for the range, and it is a reason not to propose such a criterion. These are reported as scenarios so that a reader can see the sensitivity of the range to the composition of the network. Neither is presented as a better estimate, because nothing in the analysis establishes that the excluded records are wrong.

3.8 Independence and stationarity of the annual maxima

The diagnostics support the stationarity and independence assumptions at most gauges but indicate departures at others. The lag-one autocorrelation of the annual maxima ranges from -0.344 to $+0.492$, and the Ljung–Box statistic over five lags is significant at the 5% level at three of the 13 gauges: 351006, 351007, and 351201. The Mann–Kendall diagnostic indicates a significant monotonic trend in four of the 13 series: a decrease of 2.226 mm per year at 351003 and increases of 1.307, 1.111, and 0.670 mm per year at 351004, 351007, and 351006.

Two gauges, 351006 and 351007, appear in both sets. The overlap suggests that temporal change may contribute to the serial association measured at those two gauges, although these diagnostics do not establish that the two effects are the same thing. Gauge 351003 carries the strongest trend in the network but shows no serial dependence, with a Ljung–Box p-value of 0.24, so the two diagnostics are not interchangeable.

Neither result is grounds for setting a gauge aside, and none was. What they do is qualify the reading of every return level in this paper. A significant Mann–Kendall statistic is not proof that the extreme-value process is nonstationary, and serial dependence at three of the 13 gauges does not by itself justify abandoning the stationary model. But the estimates are conditional on the record analyzed here and on the assumptions of the fitted models, and they should not be presented as assumption-free design quantities. Nonstationary frequency models provide an alternative framework for infrastructure design when temporal stability is not tenable [17]. The present analysis does not fit such

models; the reported return levels are stationary-model estimates conditional on the 1981–2014 record. The two gauges that anchor the ends of the design range in Section 3.3, 351003 at the low end and 351007 among the highest, are both among those flagged, so the qualification touches the paper's central comparison rather than its margins.

3.9 Diagnostic checks on the unusual records

The three checks of Section 2.11 were applied to all 13 records. The share of wet-day values ending in zero lies between 10.1% and 18.0% across the network, and the two lowest- S gauges lie inside that range at 11.6% and 10.6%; the share ending in five lies between 9.4% and 14.1%, with the same two gauges at 9.5% and 10.0%. A chi-square test of wet-day counts by day of the week gives p between 0.29 and 0.998 at every gauge, with the two lowest- S gauges at 0.997 and 0.998 and a ratio between the most and least frequent weekday of 1.03.

On the 50 days when the other 12 gauges were wettest on average, gauge 351009 recorded a median of 21.2 mm and gauge 351010 a median of 20.4 mm, exceeding 20 mm on 58% and 54% of those days. Four gauges with unremarkable S recorded lower medians on the same days, gauge 351002 the lowest at 5.3 mm.

These results are reported as measurements. Their interpretation is that they do not support the simple explanations of an unusual occurrence–intensity structure: there was no evidence of the specific artifacts examined, namely digit rounding, a weekday-linked observation routine, or a systematic failure to register network-wide rainfall events, which these records register at appreciable depths. What produced the structure cannot be determined from rainfall records alone, and no cause is attributed here. Station metadata on instrument type, exposure, and siting history would be needed, and were not available.

3.10 Implications for design practice

Three points follow for the use of these records. First, a single provincial design rainfall carries a median error of about a quarter at 100 years and leaves some sites below their local estimate by more than half; where a gauge is available near a site, it should be preferred to a provincial aggregate. Second, the uncertainty interval should be reported alongside the point estimate. A 100-year value from 34 years of record is an extrapolation to a return period nearly three times the record length, and Equation (11) puts its resampling spread above 100% at seven of the 13 gauges; a design value quoted without an interval overstates what the record can support at those sites, and changing among the three candidate distributions does not resolve the dominant long-return-period spread, whereas additional record length would directly address the limited information available for tail estimation, and regional frequency analysis [18] could be evaluated as a way of increasing the effective sample size, subject to the homogeneity of the regional rainfall regime. Third, an unusual record is not by itself grounds for exclusion. The gauges with the most unusual occurrence–intensity structure in this network passed all three diagnostic checks, and removing them changed the network range without any evidence that it improved the estimates.

3.11 Limitations

The network is 13 gauges over a province with substantial relief, so the between-gauge range rests on sparse spatial support and no interpolated surface was produced. Thirty-four years is short for estimating a 100-year quantile, which Section 3.5 quantifies rather than resolves. Only the annual maximum daily depth was analyzed; other durations relevant to small-structure design were not, and the results should not be transferred to them. Three distributions were compared, not the wider set available [9], [12], so Equation (13) measures the effect of choosing among these three. Both bootstraps resample annual maxima independently and therefore do not preserve serial dependence, so the intervals at the three gauges with significant Ljung–Box statistics should be read with more caution than the rest; a block resampling scheme would be the way to relax that assumption. The intervals should also be read in light of known limitations of nonparametric resampling in extreme-value applications, where performance is sensitive to sample size and uncertainty can be understated at samples of the order used here [19]. The fixed-selection bootstrap conditions on the distribution chosen from the original sample and therefore does not propagate model-selection uncertainty; the selection-inclusive bootstrap addresses this by reselecting the distribution within each replicate, and Section 3.5 reports how much the two differ. The associations of Section 3.4 comprise 39 rank correlations reported without adjustment for multiple comparisons, and are read as a pattern across return periods rather than through individual p -values. The associations of Section 3.4 rest on 13 gauges and are descriptive. The return levels are stationary-model estimates conditional on the 1981–2014 record, and Section 3.8 reports where that condition is least comfortable.

4 CONCLUSIONS

Four findings follow from the analysis of 13 gauge records across Uttaradit province in northern Thailand.

Spatial variation among gauges was larger than the sensitivity of the estimates to the choice among candidate distributions, and the gap widened with return period: the between-gauge ratio rose from 3.83 at 2 years to 12.53 at 100 years while the within-gauge spread across the three candidates reached a median factor of 1.56 at 100 years.

The occurrence–intensity structure was more strongly associated with shorter return-period estimates than with the upper tail. Mean wet-day intensity correlated with the design estimate at 0.73 at 2 years and 0.48 at 100 years, and neither it nor the wet-day count was associated with the width of the interval or with the fitted shape parameter.

Long-return-period estimates were weakly constrained by a 34-year record, and model reselection added materially to that spread at some gauges but not at others. At 100 years the median relative interval width was 145.9% with model selection held fixed and 217.7% when the model was reselected within each replicate. Across gauges, the ratio of the selection-inclusive width to the fixed-selection width ranged from 0.71 to 10.25. The widest increases occurred where the selection was least firm, and 11 of the 13 gauges had a second-ranked distribution within 2 AICc units of the one selected.

A single provincial value produced substantial site-level representation error. The network mean at 100 years fell below the local estimate at five gauges, by as much as 65%, and exceeded it at eight, and no alternative provincial statistic removed both.

These findings are conditional on the 1981–2014 record, on a stationary model formulation, on the three candidate distributions compared, and on bootstrap resampling that treats annual maxima as independent. Independence and stationarity diagnostics identified departures at some gauges, so the long-return-period estimates should not be read as assumption-free design quantities. For practice this argues for using a local record where one exists, for quoting an interval alongside any long return period, and against treating an unusual record as grounds for exclusion: the most unusual records in this network showed no evidence of the specific artifacts examined, and setting them aside narrowed the reported range without evidence that it improved any estimate.

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6 AUTHOR CONTRIBUTIONS

S.P.: conceptualization, methodology, software, formal analysis, data curation, visualization, writing—original draft, writing—review and editing. The author has read and agreed to the published version of the manuscript.

7 CONFLICTS OF INTEREST

The author declares no conflict of interest.

8 DECLARATION OF GENERATIVE AI

During the preparation of this manuscript the author used a generative artificial intelligence assistant for code review and language editing. The analysis was executed by documented code, and all computational outputs and scientific interpretations were independently verified by the author, who takes full responsibility for the publication.

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